

**West Bengal State Electricity Distribution
Company Limited**

(A Govt. Of West Bengal Enterprise)



**TECHNICAL SPECIFICATION FOR 12kV, 800A, 18.4kA,
MULTI PANEL (Incoming, Bus Coupler with Adaptor
Panel and Outgoing) SHUNT TRIP, INDOOR TYPE SCADA
COMPATIBLE SWITCHGEAR WITH MOTOR OPERATED
SPRING CLOSING VACUUM CIRCUIT BREAKER**

April'2024

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Page 1 of 70

**1.0 CODES&STANDARDS:**

Ratings, characteristics, tests and test procedures, etc. for the 11kV metal-clad switchgear panels and all the protection relays, measuring and indicating instruments and the control and monitoring devices and accessories, including current transformers and voltage transformers shall comply with the provisions and requirements of the standards of the International Electrotechnical Commission (IEC), IS where specified.

The latest revision or edition in effect at the time of bid invitation shall apply. Where references are given to numbers in the old numbering scheme from IEC it shall be taken to be the equivalent number in the new five-digit number scheme. The bidder shall specifically state the precise standard, complete with identification number, to which the various equipment and materials are manufactured and tested. The bid document may not contain a full list of standards to be used, as they only are referred to where useful for clarification of the text.

Standard Name / No	Standard's Description
Indian Electricity Rules 1956	Latest edition
Indian electricity act 2003	Latest edition
Switchgear and control gear	IEC: 60694, IEC: 60298, IEC: 62271-200, IEC: 60529, IS: 3427, IS 12729, IS 12063, IS:13947, IS: 9046
Circuit Breaker	IEC 62271-100, IS 13118, IS 2516
Vacuum Interrupter	IEC 60056
Current Transformers	IEC:60185, IS 16227 (PART 1 & 2)
Voltage Transformer	IEC:60186, IS 16227 (PART 1 & 3)
Indicating Instruments	IS:1248
Energy Meters	As mentioned in Technical Specification of Energy Meter
Relays	IS 8686, IS 3231, IS 3842
Control switches and push buttons	IS 6875
Electromagnetic Compatibility	IEC 61000
Arrangement of switchgear bus bars, main connection and auxiliary wiring	IS 375
Code of practice for phosphating iron & steel	IS 6005
Colours for ready mixed paints	IS 5

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designed life time. Particularly the power supply modules of the protection and control devices shall be designed for minimum heat generation and effective heat dissipation to ensure that the temperature of these devices enclosed in the relay panels at the above listed ambient temperatures shall not exceed the maximum operating temperature of the device.

3.1 Tropicalization

(a) All equipment must be designed for operations in the severe tropic climate conditions and fully comply with climatic aging tests as per IEC 60932-class 2.

In choosing materials their finishes, due regard shall be given to the humid tropical conditions under which the switchgear will be called upon to work. The manufacturer shall submit details of his usual practice which have proven satisfactory for application to the parts of the Switchgear panels, which may be affected by tropical conditions.

(i) Metals:

Iron and Steel are generally to be painted or galvanized as appropriate. Indoor parts may alternatively have chromium or copper-nickel plates or other approved protective finish.

Small iron and steel parts (other than rustless steel) of all instruments and electrical equipment, the cores of electromagnets and the metal parts of relays and mechanisms shall be treated in an appropriate manner to prevent rusting.

(ii) Screws, Nuts, Springs, e.t.c.:

The use of iron and steels shall be avoided in instruments and electrical relays wherever possible. Steel screws shall be zinc, cadmium or chromium plated or where plating is not possible owing to tolerance limitations, shall be of corrosion resisting steel. Instrument screws (except those forming part of a magnetic circuit) shall be of brass or bronze. Springs shall be of non-rusting material, e.g., phosphor-bronze or nickel silver, as far as possible.

(iii) Rubbers:

Neoprene and similar synthetic compounds, not subject to deterioration due to the climatic conditions, shall be used for gaskets, sealing rings, diaphragms, etc.

3.2 WORKING STRESS AND EQUIPMENT / APPARATUS DESIGN

3.2.1 General

a) The design, dimensions and materials of all parts shall be such that they will not suffer damage under the most adverse conditions nor result in deflections and vibrations, which might adversely affect the operation of the equipment. Mechanisms shall be constructed to avoid sticking due to rust or corrosion.

b) The equipment and apparatus shall be designed and manufactured in the best and most substantial and workmanlike manner with materials best suited to their respective purpose and generally in accordance with up-to-date recognized standards of good



practice.

c) Whenever possible, all similar parts, including spare parts, shall be made interchangeable. Such parts shall be of the same materials and workmanship and shall be constructed to such tolerances as to enable substitution or replacement by spare parts easily and quickly.

d) All equipment shall be designed to minimise the risk of fire and consequential damage, to prevent ingress of vermin and dust and accidental contact with electrically energized or moving parts. The switchgear panels shall be capable of continuous operation with minimum attention and maintenance in the exceptionally severe conditions likely to be obtained in a tropical climate and where the switchgear is called upon to frequently interrupt fault currents on the system and also where the duty of operation is high.

3.2.2 Strength and quality

a) All steel castings and welding shall be stress-relieved by heat treatment before machining, and castings shall be stress-relieved again after repair by welding.

b) Liberal factors of safety shall be used throughout, especially in the design of all parts subject to alternating stresses or shocks.

3.2.3 Designed data for low voltage equipment

Low voltage equipment and installation shall be designed in accordance with EMC(Electromagnetic Compatibility, IEC 61000) directives. The rating and design criteria for low voltage equipment shall be as follows:

AC Supply Rating system

- i. Rated voltage between phase 415 V AC
- ii. Connection type 3ph 4wire
- iii. Rated voltage between phase to earth 240 V AC
- iv. Grounding system PME
- v. Frequency 50 HZ
- vi. Voltage variation $\pm 10 \%$
- vii. Frequency variation $\pm 5 \%$
- viii. Power frequency 1 min, Test Voltage 3 kV
- ix. Thermal rating of conductors 120 % of load

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The AC supply shall be used for power circuit and for lighting, indication, motor controls and similar small power circuits. Unless otherwise specified, the equipment provided under this tender is to be capable of reliable operation at voltages as low as 85% of the rated voltage, and to withstand continuously up to 110% supply voltage above the rated value of 240V or 415VAC.

DC Auxiliary Supply Rating

- i. Equipment/Device Rated voltage 30V DC
- ii. Connection type 2 wire.
- iii. Voltage variation 24 to 40 V DC

The auxiliary DC supply shall be used for controls, indication, alarm, protection relays, and circuit breaker tripping and closing circuit, etc. All equipment and apparatus including the circuit breakers, protective relays, control devices and accessories, measuring and indicating instruments and electronic equipment shall be capable of satisfactory operation at 80% to 130% of the rated DC supply voltage. However, in case of VCB, for tripping the range should be 70% to 110% and for closing that should be 85% to 110%.

3.2.4 Electrical controls, auxiliaries and power supplies

(a) Responsibility for electrical control and auxiliaries.

The manufacturer shall provide all control, indication, alarm and protection devices and all auxiliary equipment with wiring and interconnecting cable which are integral parts of or are directly associated with or mounted on the switchgear panels to be supplied under this tender. The design of protection and control schemes for the switchgear panels shall be subject to approval of WBSEDCL.

b) Operation and control.

Interlocking devices shall be incorporated in the control circuit to ensure safety, and proper sequence and correct operation of the equipment. The scheme will be finalised during detailed engineering and drawing approval.

3.2.5 Corona and radio interference

- a) Switchgear shall electrically be designed to avoid local corona formation and discharge likely to cause radio interference.

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b) The design of jointing of adjacent metal parts and surfaces shall be such as to prevent corrosion of the contact surfaces and to maintain good electrical contact under service conditions.

c) Particular care shall be taken during manufacture of busbar and fittings and during subsequent handling to ensure smooth surface free from abrasion. All joints on the busbar and the circuit within the switchgear board shall be silver or tin-plated to ensure good electrical connection.

4.0 PANELCONSTRUCTION

In the event of direct conflict between various order documents, the precedence of authority of documents shall be as follows –

4.1	Enclosure Type	Dead front, floor-standing, rigid welded steel frames fully compartmentalized, Metal clad, Vermin Proof, suitable for indoor installation and provision for bolting to the floor.
4.2	Enclosure degree of protection	IP 5X for High Voltage compartment and IP4X for Low Voltage compartment
4.3	Enclosure Material	CRCA steel
4.4	Load bearing members	Minimum 2.5mm thick
4.5	Doors and covers	Minimum 2.0mm thick. Floor to circuit breaker door clearance minimum 15 mm shall be maintained so that Insulation mat can be placed.
4.6	Gland Plate (detachable type)	3.0mm MS detachable type for 3 core cable and aluminium 5.0mm for single core cables. Cable compartment shall have an anti vermin guard plate for protection against entry by rats, rodents etc.
4.7	Maximum Dimension of the Panel (WxDxH)	700mmx2000mmx2700mm.
	Maximum operating height of the panel	Operating height 1800 mm (max).
4.8	Extensibility	The Switchgear shall be designed so that future units can be added to each sides (unless coupled to other equipment). A removable plate will cover any unused openings in the side of the panel.
4.9	Extension Bus and Bus Wires	In case of Single feeder panels, Identical Bus Bars and Bus Wires for connection with adjacent Incomer / Bus Coupler / Feeder needs to be supplied.
4.10	Separate compartment for	1. Bus bar, 2. circuit breaker, 3. HV incoming / outgoing cable, 4. LV instruments relays, 5. Metering and SCADA terminal at top of the panel

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4.11	Breaker compartment door	Breaker chamber door should be hinged type with suitable locking arrangement. Breaker chamber door with bolted design is not acceptable (design with breaker trolley as the front cover is not acceptable)
4.12	Breaker to bus bar compartment	Through seal off bushings/Spout
4.13	Breaker to cable	Through seal off bushings/Spout
4.14	Pressure relief devices	To be provided for each HV Compartment. Each compartment shall be separated from adjacent one by sheet steel barrier.
4.15	Bus support insulator	Non hygroscopic, track-resistant, high strength, epoxy insulators (calculation for validating dynamic force withstand capability to be submitted during detailed engineering)
4.16	Fixing arrangement i. Doors ii. Covers iii. Gasket	Concealed hinged Bolted with SS bolts and Neoprene rubber gasket, washer
4.17	Required HV cable termination height in the cable compartment and cable termination barrier	550mm for 11 kV Suitable 4 nos FRP phase Barriers made of non hygroscopic material preferably Fibre Glass sheet should be provided in cable chamber between phases and phase to earth.
4.18	Panel Base Frame	Steel base frame as per manufacturer's standard.
4.19	Handle	Removable bolted covers for cable chamber and busbar chamber shall be provided with "C" type handles
4.20	Prevention of Internal Arc	Shall be type tested against internal arc as per provision in IEC 62271-200. The Circuit Breaker, busbars and cable compartments shall be provided with arc venting outlet. The doors for the compartment shall be capable of withstanding the effects of maximum internal arcing fault without being blown off and causing danger to personnel and other equipment. This should be proven by successful testing for 18.4 KA with duration 0.1 second as per relevant IEC standard. Higher rated panel may also be accepted. In all cases manufacturer has to offer their exact type tested design panels. Breaker chamber door should be hinged type with suitable locking arrangement. Breaker chamber door with bolted design is not acceptable.

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4.21	Test Terminal Block	13 Ways, 50A, 1100V with back connection. Dia of Entries is 5 mm & overall dimension 250X50 mm, covered with insulated cover. TTB should be Disconnecting Link type suitable for 3 Phase 4 Wire Energy Meter. Screw type TTB is not acceptable.
4.22	Multiway terminal Block and low voltage wiring	Delinking type, Rail/Channel mounted, Terminal Connector to be used in CT Circuit & Screw type for other Circuit. The Terminal Blocks should be suitable for 2.5 sq.mm wire and covered with insulated transparent cover. Pitch should be minimum 8mm & 10mm for Screw type & Delinking type connectors respectively. The low voltage cable shall be enclosed in grounded metal conduit when routed through a high voltage compartment. Control wiring shall be neatly bundled and tie wrapped where applicable. Wiring shall be protected from rubbing against door flanges or other parts of the enclosure.
4.23	Cable Tray	Netted Metal cable Tray of suitable size at the rear side of Switchgear, preferably running at the top along the panel for carrying the signal cables for SCADA interface to be provided.
4.24	Remote Terminal Box	To be provided suitably at the back side of the panel for termination of the following cables : A . Main AC / DC B . Inter trip from up stream C . SCADA related terminations D . Inter connection for Differential protection etc.
4.24.1	Voltage Presence Indicator (VPI)	a. Live line Indicators: - Capacitive voltage indicators shall be provided on feeder side in outgoing and incoming feeders to indicate the voltage presence in each phase. Voltage Presence Indicator (VPI) to be placed at suitable rear/front side of the panels (Position to be finalised during drawing approval stage). b. The panel shall have Voltage Presence Indicator (VPI) to warn the operator against earthing of live connections. c. Use of voltage presence indicator sensors having high linearity over the entire working range and incorporating potential / capacitive divider circuit for connection to voltage indicator lamps.
4.25	Circuit Breaker	
4.25.1	Mounting	On withdraw able truck/trolley or carriage, with locking facility in service position. Switchgear truck/trolley should be floor mounted. Racking-in and Racking-out should be such that one person

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4.25.2	Switching duty	a) Transformer oil filled and dry type b) Under ground cable with upto 10KM.
4.25.3	Interrupting medium	Vacuum
4.25.4	Breaker operation	Three separate identical single pole units operated through the common shaft and shall be fully interchangeable both electrically and mechanically. Circuit breaker poles between the interrupters and primary plug-in contacts shall be fully insulated with durable material. Each breaker shall be provided with Mechanical 'ON' and 'OFF' facility by operating suitable closing and opening devices. each breaker shall be provided with Mechanical 'ON' and 'OFF' indicators. Each breaker shall have three positions- service, test and isolated/withdrawal marked. mechanical safety interlocks shall be provided so that it is not possible for a circuit breaker a. To be put into the cubicle unless the truck is secured in position. b. To be either draw out or draw in from and to the service position unless its contacts are safely open. c. To be withdrawn or inserted in the fixed housing unless it is at the withdrawable position. d. To be operated in service position unless its primary and secondary isolating contacts are fully engaged. e. Rack In and Rack out of the VCB trolley possible when the Breaker door in close position. The interlock should be clearly mentioned on the front door of the Breaker. f. The circuit breaker racking equipment can be padlocked in closed position. g. Electrical close/trip operation should be dependent on Local/Remote switch. However, protection trip and
4.25.5	Operating mechanism	Re-strike free, tripfree both electrically and mechanically, with electrical anti-pumping feature One O-C-O operation possible after failure of power supply to the spring charging motor. Motor wound, spring, charged, stored energy type with Manual charging facility. One no. Breaker Truck operating handle for every three panels needs to supply.
4.25.6	Trip and Closing Coil	To be rated for substation DC voltage. Suitable for operation at minimum operating voltage of 70% for tripping and 85% for closing operation. Maximum Burden shall be about 250 watt for each coil.



4.25.7	Truck	The circuit breaker shall be mounted on an inbuilt carriage to facilitate isolation and withdrawal of the breaker. Trolley of the VCB should be horizontally isolated and there should be a minimum ground clearance of 0.55 meter between thimble point of cable termination and GL. There should be a minimum clearance of 300 mm between any 11kV live bus and GL. Trolley should be detachable from panel without any additional supporting trolley i.e. VCB trolley itself move on ground level.
4.25.8	Adaptor Panel	Separate Adaptor Panel with each Bus Coupler panel shall be designed.
4.26	Breaker Indications and push buttons	
4.26.1	ON/OFF/Emergency trip push button	i) Electrical and Mechanical type in front of the panel ii) Mushroom type Emergency Off push button will be provided with a protective flap in front of the panel.
4.26.2	Mechanical ON-OFF indication	On breaker trolley front
4.26.3	Operation counter	On breaker trolley front
4.26.4	Test-service position indicator	On the front of the panel
4.26.5	Mechanism charge/discharge indicator	On breaker trolley front
4.26.6	Breaker positions	Service, test and isolated
4.26.7	Inter changeability	The Circuit Breakers of Incomer, Bus Coupler & Feeder Panels should be interchangeable.
4.26.8	Breaker control	On panel front only
4.26.9	Handle	Breaker shall be provided with handles for easy handling, rack in-out operation and manual spring charging as applicable.
4.27	Functional Requirements	
4.27.0	Interlock and safety devices	
4.27.1	Breaker compartment door opening	Cannot be opened unless breaker is OFF and racked out to test/ isolated position
4.27.2	Breaker compartment door closing	Should be possible even when breaker in isolated position
4.27.3	Racking mechanism safety interlock	Mechanical type
4.27.4	Racking in or out of breaker inhibited	When the breaker is closed
4.27.5	Racking in the circuit breaker inhibited	Unless the control plug is fully engaged
4.27.6	Disconnection of control plug inhibited	As long as the breaker is in service position

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23/04/24

23/04/24

23/04/24

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Page 12 of 70

Technical Specification for 11 kV SCADA compatible Shunt trip switchgear panel

 23/04/24
 23/04/24
 23/04/24
 23/04/24
 23/04/24
 23/04/24
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6.1.0	Type	Potential Transformer should be Three Phase Five Limb type or it should be combination of three number Single Phase PT housed in a withdraw able carriage. Shall be cast resin type with insulation class of E or better. Service position locking mechanism shall be provided and indicated by bidder in relevant drawing. Rigidity of primary star point with earth bus in service position shall be confirmed. Connection of Primary neutral with main earth bus to be made with solid copper of minimum 50 sq mm. Contact tips of primary/secondary contacts shall be silver plated. Correct polarity shall be distinctly marked on primary and secondary terminal. Secondary terminal studs shall be provided with at least three nuts, two plain and two spring washers for fixing leads. The stud, nuts and washers shall be of brass, duly nickel plated. The minimum outside diameter of the studs shall be 6 mm. the length of at least 15 mm shall be available on the studs for inserting the leads. The space clearance between nuts on adjacent studs when fitted shall be at least 10 mm.
6.2.0	Mounting	Rail mounted on top of the Unit and connected on bus side. It can be plugged into and withdraw from service by pulling or pushing the PT by the handle provided on the PT. This action traverses the PT along the rails and shall automatically operate the spout shutters. The shutter drive also forms a latch which holds the PT in the service position and this latch shall be required to be released before PT can be isolated. Access to the PT and the reinforcement in the Panel for allowing a person to stand on the Top should be provided. Locking arrangement of the PT with the carriage and secondary fuses are to be provided. Mounting of PT on the breaker truck is not acceptable.
7.0.0	Equipment Earthing	
7.1	Material of Earthing bus	Two separate earthing terminals shall be provided in each panel and shall be connected to the earth bus within the panel. The earth bus shall be of tin plated copper and shall have adequate cross sectional area. Earthing conductors shall be of annealed high conductivity stranded Copper in accordance with Table-4 in BS.6346 and protected with an extruded PVC sheath of 1100 volt grade. The earthing conductor on the primary equipment as well as for external connection to substation earthing grid shall be adequate to carry the rated switchgear short-circuit current of 18.4kA for 1 second.
7.2.	Earth bus joints	A ground bus rated to carry maximum fault current shall be furnished along the full length of the panel board. Each stationary unit shall be grounded directly to ground bus. All bolted joints in the bus will be effected by connection of two bolts.

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7.3.	Size	Rectangular Earth Bus of Minimum size 25mmX6mm
7.4.	Enclosure and non-current carrying part of the switchboard/components	Effectively bonded to the earth bus
7.5.	Hinged doors	Earthed through flexible copper braid
7.6.	Circuit breaker frame / carriage	Earthed before the main circuit breaker contacts/control circuit contacts are plugged in the associated stationary contacts.
7.7.	Metallic cases of relays, instruments and other LT panel mounted equipment	Connected to the earth bus by independent copper wires of size not less than 2.5 sq mm with green colour insulation. For this purpose LT compartment should have a clear designated earth bus to which earth connections from all components are to be connected.
7.8.	CT & PT Secondary neutral	Earthed at one place at the terminal blocks through
8.0	Indicating Meters	Flush Mounted, back connected, dust proof with Industrial grade "A" classification and conforms to IS:1248(1968)
8.1	CT operated Ammeter	3 no. Digital Ammeters to be provided in Incomer and Feeder panels for measuring load current of each phases. Display : 3 ½ -digit display, Aux. Power: : 230 V AC CTR: 600/1A for Incomer and 200/1A for Feeders.(Site Selectable) Accuracy Class: 1.0 or less. Size : 96 X 48 mm.
8.2.	Direct Ammeter	1 nos. Separate Digital Ammeter to be provided for measuring heater current drawal in all the Panels. Display : 3 ½ -digit display, Aux. Power: : 230 V AC Accuracy Class : 1.0 or less. Size : 96 X 48 mm.
8.3.	P T operated Voltmeter	One no. Digital Voltmeter to be provided in Incomer Panels for measurement of Phase to Phase system voltage, along with a Voltage Selector Switch. Display : 3 ½ -digit display, Aux. Power: : 230 V AC PTR : 11000 / 110 V Accuracy Class : 1.0 or less. Size : 96 X 48 mm.
8.4.	Direct Voltmeter	One no Analog DC Voltmeter to be provided in the Bus Coupler panel for measurement of Control DC voltage.

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12.8	Push button for Annunciator accept and reset	Two nos Push button, one for annunciator accept and one for annunciator reset to be wired upto annunciator.
13.0	Internal wiring	1100/750 V grade PVC insulated stranded flexible (FRLS) copper wire.
13.1.	Wire Size	2.5 sq. mm for CT circuit, 1.5 sq mm for PT and 1.5 sq. mm for control circuit, 4.0 sq. mm for Main, AC & DC and 2.5 sq. mm for others Bus wiring. i. A suitable wiring duct system firmly fixed on the panel and having covers shall be installed for front to rear and inter panel wiring to provide easy access for inspection and replacement of the wires. ii. Wiring between terminals of the various devices shall be point to point. Splices or tee connection will not be acceptable. iii. Facilities for short-circuiting the current transformer secondary while the switchgear is in service shall be provided. iv. Wires shall be suitably bunched adequately supported to prevent sagging. and it shall have sufficient clearance from High voltage system. v. Sufficient Bus wires shall be provided for interconnecting with adjacent units both ways and also for future connection.
13.2	Wiring Colour code	
13.2.1	PT and CT	Red Phase : Red, Yellow Phase : Yellow Blue Phase: Blue, Neutral : Black.
13.2.2	DC Circuit	Gray
13.2.3	AC Circuit	Black
13.2.4	Earth	Green
13.3	Ferrules	At both ends of wire with same stated marking. Plastic ferrules shall be used conforming IS. Interlocked type (one additional red colour ferrule for all wires in trip circuit)
13.3.1	Ferrule marking	
13.3.1.1	AC Circuit	H1, H2 ,H3
13.3.1.2	Metering CT Circuit	D11, D31, D51
13.3.1.3	O/C and E/F protection CT circuit	C11, C31, C51
13.3.1.4	REF/Differential protection CT circuit	A11, A31, A51
13.3.1.5	Main DC distribution circuit	J1, J2, J3
13.3.1.6	Control and protection circuit	K1, K2 ,K3
13.3.1.7	Indication and annunciation circuit	L1, L2 ,L3
13.3.1.8	Motor circuit	M1, M2 ,M3
13.3.1.9	PT circuit	E11, E31, E51, E71.....

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13.3.1. 10	Spare contact circuit	U1, U2, U3 .../ S1, S2, S3
13.4	Lugs	Tinned copper, pre-insulated, fork type and pin type as applicable. Only Ring type is acceptable in CT & PT Circuit.
13.5	Spare contacts of relays, timers etc	Wired up to the terminal blocks
13.6	Wiring enclosure	Plastic channels, inter panel wiring through PVC sleeves or suitable grommets.
13.7	Inter panel wiring	Wiring with ferrule to be terminated in the adjacent shipping section will be supplied with one end terminated and the other end bunched and coiled.
14.0	Terminal Blocks (TB)	
14.1.	Terminal Blocks(TB)	1100V grade, 10amps min. rating, Nylon 66, screw type suitable for 2 nos. Lead.
14.2.	Terminal for CT & PT secondary leads	With provision for shorting with screw driver operated sliding link. a) CT shorting links shall be provided to short CT circuits under live system condition. b) Isolation links shall be provided on the trip circuits, alarm and on the VT circuits to allow easy isolation without disconnecting the wires from TBs.
14.3	Spare terminals	At least 10% or 4 nos (which is higher) in each panel
14.4	TB shrouds & separators	Moulded non-inflammable plastic material
14.5	Clearance	
14.5.1	Clearance between 2 set of TB	100 mm min.
14.5.2	Clearance with cable gland Plate	250 mm min
14.5.3	Clearance between AC / DC set of TB	100 mm min
15.0	Test Terminal blocks for energy meter	13 Ways, 50A, 1100V with back connection. Dia of Entries is 5 mm & overall dimension 250X50 mm, covered with insulated cover. TTB should be suitable for 3 Phase 4 Wire Energy Meter. TTB used for CT Circuit should be Link type. Screw type is not acceptable.
16.0	HT Cable Type	3 Core XLPE, 11 kV grade, Aluminium, 300mm ²
17.0	HT Cable Termination	Cable entry from rear bottom side for Incomer and Outgoing feeders. The cable termination shall be located at least 250 mm from the CT primary terminals. For Incomer, double cable termination arrangement to be provided with two sets of nut and bolts. Copper terminator strip of adequate size shall be provided for termination of cables and shall have adequate height inside to accommodate the heat shrinkable type indoor cable termination.

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18.0	Multi-core cable termination box-- Purpose	Incoming Panel- External DC supply from DCDB, External AC supply from ACDB, Remote Electrical Close and trip operation, Inter trip operation from upstream 33 kV circuit, Breaker spare aux. Contacts, Spare contacts of L/R Switch and Spring Charge Limit Switch etc. Feeder Panel -- Remote electrical close/trip operation, remote indication for close, trip, spring charged and trip ckt supervision, status of L/R switch, Local annunciation for remote close and trip operation, Breaker spare aux. Contacts. Bus Coupler -- External DC supply from DCDB, External AC supply from ACDB, Remote Electrical Close and trip operation, Breaker spare aux. Contact. External DC and AC supply arrangement to be provided in Incomer and Bus Coupler panel only in a switch board.
19.0	Operation Counter	Each breaker shall be provided with operation counter.
20.0	Vacuum Interrupter	Capable to withstand minimum 100 full short circuit operations at 26.3 kA as per test duty 1 to 5 of IEC-56. The continuous Current rating should be more than 800A. Manufacturer's test report/literature to be supplied along with tender.
21.0	Insulated spacer	It shall be made of insulated fibre glass materials shall be provided in High voltage system, to protect from absorption of moisture present in the air.
22.0	PT selection scheme	Each Incoming and outgoing breaker panel shall be provided with suitable manually operated selector switch having break before make type contacts including necessary wiring for selection of voltage from different PT secondary's available for connection with energy meters and numerical relay for Incomer panel. One output of the numerical relay to be configured as "PT under voltage" for one of the annunciator window as Non Trip indication. Each panel should be provided with 3 PT scheme and current rating of the PT Selector Switch should be minimum 16 Amp. Voltmeter of incoming breaker panel shall be connected directly from respective PT secondary of associated unit.
23.0	Relays for protection and control	
23.1	General Features	
23.1.1	Relay type	Numerical with self monitoring features
23.1.2	Mounting	Flush mounted IP 5X with key pad on front

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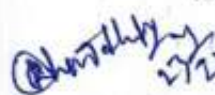
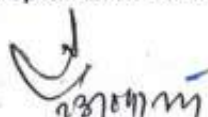
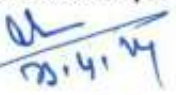
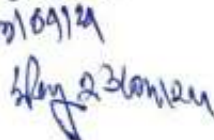
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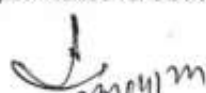
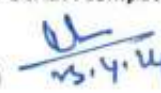
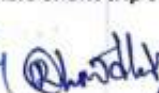
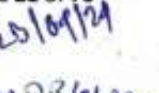


23.1.3	Relay characteristics	Numerical IED with Multiple characteristics like IDMT, DMT, instantaneous with compatible choices of time delays and multiple settings for multiple functions like overcurrent, earth fault etc. along with control of breaker, measurement and status etc. As per detailed technical specification attached in a separate sheet (Annexure-E).
23.1.4	Relay communication	With IEC 61850 protocol by native two nos. RJ45 /F.O, 100 MBPS communication ports. Separate front port (USB/RJ45) for Relay parameterisation.
23.1.5	Relay input signal	From CT, with auxiliary DC supply
23.1.6	Relay terminals	Shall be screw type terminals large enough to accommodate 2.5 sq. mm. cables and shall be located at the back of the Relay.
23.1.7	Relay contacts	Shall make firmly without bounce and the relay mechanism shall not be affected by panel vibration or external magnetic field.
23.1.8	Electromagnetic Compatibility	Relay thermal rating shall be such that the fault clearance times on any combination of current and time multiplier settings shall not exceed the thermal withstand capability of the relay. Compliant to EMC directives as per IEC.
23.1.9	Relays for auxiliary, supervision, trip and timer relays	Static or attracted armature type with short pickup time of less than 30 ms.
23.1.10	Relay reset	Self reset contacts except for Master Trip relays.
23.1.11	Operation indicators	With hand reset operation indicators (flags) or LED with Push buttons for resetting, for analyzing the cause of operation
23.2	Master Trip relay	High Impedance and High speed relay flush mounted having coil cut-off contact with at least 4NO and 4NC contact and electrical reset facility capable to make, carry and break trip coil current of Circuit Breaker and capable for future integration with SCADA. It should be immune to capacitance discharge currents and leakage current. Operating time should be less than 20ms. Terminals shall be screw type to accommodate 1.5 sq. mm. Cable and located at the back of the relay. Terminals shall be clearly marked. Contact configuration shall be drawn on the relay casing.
23.3	Fault recording	Relay shall have the facility for recording of various parameters during a fault. It should be possible to set the duration of record through settable pre fault and post fault time. It should be possible to down load the data locally or from SCADA remotely.
23.4	Auxiliary supply	Operate on available 30 V. DC supply. To reduce the effect of Electrolysis, relay coil shall be so connected such that they are not continuously connected from the positive pole of the station battery.
23.5	Test facility	In built.

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23.6	Protection in Incomer & Outgoing panel	3 phase overcurrent protection & Earth fault protection
23.7	Additional requirement	
23.7.1	For each breaker panel	To be provided with anti pumping relay/contactor (94), (plug-in type contactor will not be accepted)
23.7.2	Auxiliary relays, coupling relays, transducers etc.	To effect interlocks and to exchange signals of status & Control from remote.
24.0	Additional Requirement	The schematic diagram of control and indication circuit on a durable sticker shall be fixed in a suitable place in Incomer and feeder panels and Annunciation scheme in the Bus Coupler panel. The wiring arrangement, position of fuse, TBs etc, in the panel should be such that each and every connection can be accessed easily. Fuse tops can be opened /inserted easily. Only ring type sockets should be used for wire termination in TB/ Relays/ Meters/TTB etc. Fork type socket can be used in Annunciator only. Separate switch to be used for cubicle light instead of Door switch. Circuit Label incorporating identification, information of incoming/outgoing/bus coupler breaker shall be provided. Connection diagram plates for the PT and CT shall be provided as per provision in the relevant IS. All name, rating , circuit label, connection diagram plate shall be fixed with screws or by riveting instead of fixing by adhesive.
25.0	Annunciation Scheme	6Window shall be provided in each incomer and outgoing panel, 8 window for Bus Coupler panel. It should be Microprocessor based having inbuilt Accept/Reset/Test/Mute push buttons for trip and non-trip functions. DC operated common Hooters shall be provided in Bus Coupler breaker panel. It shall have provision of inbuilt watch Dog and Fast fault indication with Trip (O/C , E/F and other trip signals if any)and non-trip alarm i.e AC supply fail, main DC supply fail at Bus Coupler. The functional details and scheme will be decided during finalisation of drawings.
26.0	Space Heaters	
26.1	Space heaters	60 W, 230 V AC each with timer controlled and switch for isolation. A Thermostat control unit with variable temperature control setting shall be installed to control the heater. The 240 V AC supply for the heater shall be controlled by a suitably rated single pole miniature circuit breaker.
26.2	Space heater location	Two nos. each in Breaker & HV cable compartment to be mounted on an insulator.

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26.3	Space heater Monitoring	One AC Ammeter with 0-1.0Amp range shall be provided in series with the heaters to monitor the current drawal of the Heaters. Size :96x48mm.
27.0	Switch and sockets	
27.1.	Illumination Lamp	For LV ,cable chamber & energy meter
27.2.	240V AC, 15A /5A Switch & socket	In LV chamber
28.0	Name Plates and rating plate of switchgear	
28.1	Material	Material shall be stainless steel/anodized aluminium and shall not be deformed under the service condition. The entries shall be indelibly marked by engraving with black letter on white background or vice-versa as specified and 3mm thick and 100mmX150mm(approx).
28.2	Lettering	Engraved, with white letters on black background, for use inside the control compartment, a white label, engraved in Black Letters and numbers shall be used. Each major equipment shall be provided with a rating plate containing the required information as specified in the relevant IEC standard.
28.3	Name plate for feeder description & number	On front and rear side of each panel giving feeder description. On the rear side the name plate shall be fixed on the back cover.
28.4	Identification plates/labels at each cubicle and each instrument	Approved design (not stickers)
28.5	Manufacturer name plate	On front top door of panel
29.0	Rating Plate of Switchgear	Name of the Manufacturer and Month/Year of Manufacture P.O No. and Date Type, Designation and Serial No. Rated Voltage and Current Lightning impulse withstand voltage 1 minute Power frequency withstand voltage Rated frequency Rated Current Breaking Capacity Short time current making capacity Operating sequence Rated voltage of closing and opening coil Rated voltage of spring charging motor



		Make and type of VI
		Property Label "Property of WBSEDCL"
		Guarantee for Five Years
30.0	General	a) Materials shall be new; the best quality of their respective kinds and such as are usual and suitable for work of like character. All materials shall comply with the latest issues of the specified standard unless otherwise specified or permitted by WBSEDCL. b) Workmanship shall be of the highest class throughout to ensure reliable and vibrations free operations. The design, dimensions and materials of all parts shall be such that the stresses to which they may be subjected shall not cause distortion, undue wear, or damage under the most severe conditions encountered in service. c) All parts shall conform to the dimensions shown and shall be built in accordance with approved drawings. All joints, datum surfaces and meeting components shall be machined and all castings shall be spot faced for nuts. All machined finishes shall be shown on the drawings. All screw, bolts, studs and nuts and threads for pipe shall conform to the latest standards of the International Organization for Standardization covering these components and shall all conform to the standards for metric sizes. d) All materials and works that have cracks, flaws or other defects or inferior workmanship will be rejected WBSEDCL. e) Each of the Switchgear panel shall be of unitised construction with all necessary accessories like end covers etc. However the design shall allow for extension on both sides without limit. Busbar design shall be such that panel to panel interconnection can be carried out without difficulty as and when required. In case of Single feeder panels, Identical Bus Bars for connection with adjacent Incomer / Bus Coupler / Feeder needs to be supplied. f) Explosion vents of suitable design shall be provided on the roof sheet of the busbar/cable/CT's chambers so as to enable discharge of explosive gases from inside during a flashover. g) All primary current carrying conductor bar shall be CU made and minimum current carrying capacity shall be as per CB rating if otherwise not specified. Current density to be considered not more than 1.6 A/sqmm.
30.1	Assembly	Necessary items of equipment shall be assembled in the factory prior to shipment and routine tests shall be performed by the manufacturer as per the requirements of the latest issue of IEC as specified under each equipment in these specifications to demonstrate to the satisfaction of the WBSEDCL that the switchgear panels comply with the requirements of the relevant IEC standards.

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30.2	Casting	Casting shall be true to pattern, of workmanlike finish and of uniform quality and condition, free from blowholes, porosity, hard spots, shrinkage defects, cracks or other injurious defects, shall be satisfactorily cleaned for their intended purpose.
30.3	Welding	Wherever welding is specified or permitted, a welding process, including stress relieve treatment as required if necessary, conforming to an appropriate and widely recognized professional standard shall be used. All welders and welding operators shall be fully qualified by such a standard.
30.4	Operational Details	Instructions shall be engraved on the switchgear panel, on the circuit breaker compartment describing in simple steps how to carry out correct and safe isolation, racking-in and racking-out switching operations on the circuit breaker. Similar details should be provided for the operation of the earth switch.
31.0	Surface treatment & painting	
31.1	Surface treatment	Sand blasting or by seven tank process
31.2	Paint type	Powder coated. Pure polyester base grade A structure finish
31.3	Paint shade	RAL 7032 for external & internal surface
31.4	Paint thickness	Minimum 60microns
32.0	Type Test	
32.1	Type Tests	The equipment offered in the tender should have been successfully type tested at NABL accredited third party laboratories in India or equivalent STL accredited international laboratories in line with the relevant IEC/Indian standards and technical specification, The bidder shall be required to submit complete set of the type test reports along with the offer.
32.2	Type test report validity period	Validity of type test shall be as per latest CEA guidelines, i.e., 5 (five) years from the date of offer.

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32.3	Type test report to be submitted along with the bid	A . Switchgear Panel (with circuit breaker installed) a. Lightning Impulse Voltage withstand Test b. H.V. dry 1 min power frequency withstand test c. Short time and peak withstand current test d. Short circuit test with basic duties e . Single phase breaking capacity test. f. Capacitive current switching test confirming to relevant IEC:- i) Line charging breaking current test: 10A ii) Cable charging breaking current test: 25A iii) Single capacitor bank breaking current test: 400A g. Temperature Rise test h. IP Test i. Internal Arc Test as per IEC 62271-200 j. Horizontal Acceleration due to seismic force (if available) B. Circuit Breaker a. Mechanical Endurance Test as per M2 Class of IEC b. Class E2 circuit-breakers intended for use without auto-reclosing duty as per clause no 6.112.1 of IEC. Basic short-circuit test series shall consist of test-duties T10, T30, T60, T100s and T100a as per clause no 6.106 of IEC and all sequence to be done without any maintenance in between the test and test reports to be made in single occurrence. C. Current Transformer a. Short Time Current Test b. Impulse Voltage Withstand Test c. Temperature Rise Test D. Potential Transformer a. Impulse Voltage Withstand Test b. Temperature Rise Test Copies of test certificates in respect of following bought out items:- a. Vacuum Interrupter. b. Insulators c. Bus Bar Material d. Terminal connectors e. KEMA / CPRI Certification for relay/IED i.r.o. IEC61850 compliance Note : All the type test report on Switchgear Panel & Circuit Breaker to be conducted with offered Vacuum Interrupter.
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Technical Specification for 11 kV SCADA compatible Shunt trip switchgear panel

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Page 26 of 70
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36.0	Drawing and documentation	Bidder shall submit following drawing/documents along with the bid:- 1. GA of indoor requisite set of panel Switchgear showing constructional features and space required in the front for withdrawal of breaker truck and in back, other accessories, power and control cable entry with plan elevation and views. 2. Sectional view of incomer, bus coupler & feeder panels with parts list. 3. GA of Circuit Breaker truck. 4. GA of Current Transformer 5. GA of Potential Transformer. 6. Technical particulars of Switchgears and brochures for technical data sheet of vacuum interrupter 7. Relay and device list (Bill of Quantity) with complete details. 8. Technical manual for Relays, Meters, Switches, Instrument transformer, Terminal block and Test Terminal Block. 9. Four copies of drawing, data and manuals containing above shall be submitted for approval and afterwards for final distribution. 10. Two sets of manual, leaflet and drawing for multi panel board shall be submitted separately to the C.E(Testing), Distribution Testing Deptt., WBSEDCL. Successful tenderer shall furnish all above drawings and following additional drawings for approval before commencement of supply:- i. Foundation details for requisite set of Panel Switchgear. ii. Equipment door layout for incomer, bus coupler & feeder panels. iii. Schematic Diagram for incomer bus coupler & feeder section of Switchgear iv. Protection Circuit, Metering circuit, DC control circuit for incomer bus coupler & feeder section of Switchgear v. Annunciator and Alarm scheme. vi. P.T. supply change over scheme. ix. Terminal block details for incomer, bus coupler & feeder section. vii. Cross section view for CTs. viii. Name Plate & Connection diagram for CTs. ix. Cross section view for PTs. x. Name Plate & Connection diagram for PTs. xi. Rating Plate details of the Panel xii. Manual for installation, operation and maintenance procedure.
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37.0	Guarantee	Guarantee of the total equipment including any integral part of the equipment should be for a period of 5 (five) years from the date of last despatch.
38.0	Routine & Acceptance Test	All the switchgear panels shall be tested in accordance with the requirement of IEC 62271-200. Tests shall be carried out on the circuit breakers as per the requirement of IEC 62271-100. Current transformers and Voltage transformers shall be tested in accordance with the requirement of IEC 60044-1, IS 2705 and IEC 6044-2, IS 3156 respectively. The Protection Relay shall be tested in accordance with IEC 60255 & Energy Meter shall be tested as per relevant IS. All routine tests at manufacturer's works shall be carried out and Test Reports are to be submitted to CE, P&C, WBSEDCL. All Acceptance tests shall be carried out at manufacturer's works on every lot offered for inspection as per relevant IS & IEC in presence of the WBSEDCL's representatives. Selection of samples for acceptance test as well as rejection and retesting shall be guided by relevant IS. The entire cost of acceptance and routine test that to be carried out shall be treated as included in the quoted price of tendered items. Six copies of test reports duly signed by the inspecting officers, shall be submitted to the Chief Engineer, P&C Department, Bidyut Bhavan (4th floor) Salt Lake, Kol -91.
38.1	Complete Switchgear Panel test	a) Dimensional Checks b) Operational Tests c) Primary Injection Tests d) Calibration Tests on Relays and Instruments e) Power frequency Withstand Test f) Insulation Test g) Contact resistance test of Primary joints h) Power frequency Withstand Test on secondary Wiring
38.2	Circuit Breaker test (six copies of test report to be submitted with Offer)	Routine tests. a) Operation test. b) Dielectric Test on the main circuit. c) Dielectric tests on controls and auxiliary circuits. d) Measurement of resistance of the main circuit. e) Timing test f) Design and visual checks g) Test Certificate for all resin cast/moulded components of circuit breakers whose partial discharge measurement are specified in the relevant IEC (No. 62271-100, Clause No.6.2.9)
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38.4	Current Transformer Test (six copies of test report to be submitted with Offer)	Routine tests shall be carried out at the manufacturer's plant as per the requirement of IEC 60044-1, IS 16227 (PART 1 & 2) as listed below: i. Polarity test and verification of terminal markings test ii. Ratio and phase angle error test (accuracy class; composite error test) iii. Power frequency tests on primary and secondary windings iv. Power frequency withstand tests between sections (windings) v. Inter-turn over voltage tests vi. Partial discharge measurement
	Voltage Transformer test (six copies of test report to be submitted with Offer)	Routine tests shall be carried out at the manufacturer's plant as per the requirement of IEC 60044-2, IS 16227 (PART 1 & 3) as listed below:- (a) Polarity tests and verification of terminals (b) Power frequency withstand tests on Primary windings (c) Power frequency withstand tests on secondary windings (d) Power frequency withstand tests between sections (e) Determination of errors (f) Partial discharges measurement
	Protection Relay Tests (six copies of test report to be submitted before with Offer)	a. Relay Pick-up test for all functions and phases b. Relay timing test for all functions and phases c. Conformance Test as per IEC 61850

Table-1

LIST OF LOOSE MATERIAL				
SI.NO	Object description	INCOMER &	OUTGOING PANEL	BUS COUPLER PANEL
		Qty	Qty	Qty
1	BUS BAR (AS PER DESIGN & APPROVED DRAWINGS)	1 SET	1 SET	1 SET
2	RACK IN/RACK OUT HANDLE	1 NO	1 NO	1 NO
3	PANEL COUPLING HARDWARE KIT	1 SET	1 SET	1 SET
4	HT HARDWARE FOR BUS BAR	1 SET	1 SET	1 SET
5	FOUNDATION BOLT	AS PER DESIGN & APPROVED DRAWINGS	AS PER DESIGN & APPROVED DRAWINGS	AS PER DESIGN & APPROVED DRAWINGS
6	BUS BAR SHROUDS	AS PER DESIGN & APPROVED DRAWINGS	AS PER DESIGN & APPROVED DRAWINGS	AS PER DESIGN & APPROVED DRAWINGS
7	END COVER	1	-	-
8	END PANEL GROMMET	1	-	-
9	SHORTING EARTHBAR	1	-	1

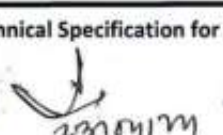
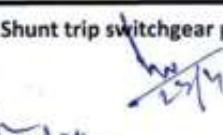
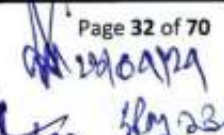
ANNEXURE -A

MANDATORY TECHNICAL PARTICULARS

1.0	Switchgear	
1.1	Type	Metal clad, air insulated with VCB type circuit Breaker
1.2	Service	Indoor
1.3	Mounting	Freestanding, floor mounted
1.4	System voltage	11 kV
1.5	Highest System voltage	12 kV
1.6	Frequency	50HZ, $\pm 3\%$
1.7	Phase	3
1.8	Rated current @ 50 DEGC ambient	800 Amp.
1.9	Short time rating for 3 sec.	18.4 kA
1.10	Insulation level (PF rms / impulse peak)	28 / 75 kV
1.11	System ground	Solidly earthed
1.12	Enclosure degree of protection	IP 5X for high and low voltage compartment
1.13	Bus bar – Main @ 50 DEGC ambient	1600A, Current Density 1.6 Amp/sq.mm.(Max)
1.14	Material	electrolytic copper, tinned /Silver plated
1.15	Bus Bar sleeve	Full voltage sleeved with shrouds on joints
1.16	Bus identification	Colour coded (R-Y-B)
1.17	Bus end connection	To be capable to safely withstand stress due to max. Short circuit current and thermal expansion. Necessary provision to be made for testing current transformer primary by removing insulated portion without difficulty.
1.18	Temperature rise above Ambient	40DEGC for conventional joints, 55DEGC for silver plated joints
1.19	Riser Bus	electrolytic copper, tinned /Silver plated, 800A, Current Density 1.6 Amp/sq.mm.(Max)
1.20	For the design and erection of Busbars the following minimum clearance shall be observed	
1.21	Phase to Phase(mm) (without Insulating Sleeve)	110 mm (approx)
1.21.1	Phase to earth(mm) (without Insulating Sleeve)	90 mm(approx)
1.21.2	Minimum nominal creepage distance	19mm/kV.
1.22	Auxiliary DC Supply	30V DC shall be controlled by suitably rated miniature circuit breaker.



1.23	Auxiliary AC supply	240V AC 50 Hz shall be controlled by suitably rated miniature circuit breaker.
1.24	Hardware	GI
1.25	Earth bus	Electrolytic Copper, Rectangular shape, 25X6mm(Min) and to carry 18.4 KA for 1 Second.
1.26	Power and control cable entry	From bottom for Power Cable. For control cables through Remote Terminal Box installed at the backside of the Panel.
2.0	Circuit Breaker	
2.1	Type of isolation & Draw out	Horizontal isolation & Horizontal Draw out.
2.2	Voltage class, insulation level, short time rating	As specified for switchgear
2.3	Rated current	800 Amps.
2.4	Duty cycle	O – 0.3sec –CO – 3min –CO
2.5	Short circuit rating	
2.5.1	AC sym. Breaking current	26.3 KA (for Vacuum Interrupter)
2.5.2	Short circuit making current	65.75KA (for Vacuum Interrupter)
2.6	Operating time	
2.6.1	Break time	Less than 80 ms.
2.6.2	Make time	Less than 100 ms.
2.7.	Range of auxiliary voltage	
2.7.1	Closing	85% - 110%
2.7.2	Tripping	70% - 110%
2.7.3	Spring charging	85% - 110%
2.8	Auxiliary Switch	Properly rated and robust in nature shall be provided in all panels and wired upto SIC (Secondary Isolated Contact) with identical SIC and ferrule numbers and sufficient number of NO and NC contacts for the following:- CB 'ON' for Local indication –NO contact CB 'ON' for Remote indication—NO contact CB 'OFF' for Local Indication — NC Contact CB 'OFF' for Remote Indication – NC Contact CB 'Auto Trip' for Local and Remote Indication—NC Contact. Other Contacts used as per scheme for closing and tripping and trip ckt supervision.
2.8.1	No. of spare contacts in Aux. Switch for WBSEDCL use	Minimum 6 NO + 6 NC
2.9	No. of spare contacts of service and test position limit switch contact	2 NO+2NC
2.10	No. of spare contacts of Spring charge limit switch	2 NO + 2 NC
3.0	Current Transformers for Incoming Panel	
3.1.	Voltage class, insulation level	As specified for switchgear

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Technical Specification for 11 kV SCADA compatible Shunt trip switchgear panel

4.11	Name Plate of CT indelibly marked	Name plate for the current transformer shall be provided with all the required details as per IEC 60044-1, IS 2705 Standards, including :- i) The manufacturer's name ii) Serial number and a type designation iii) Rated primary and secondary current iv) Rated frequency v) Rated output and corresponding accuracy class for each secondary winding, including the rated accuracy limit factor and Instrument security factor for protection and metering secondary windings respectively. vi) The highest voltage of the equipment vii) The rated Insulation level viii) The rated short-time thermal current (<i>I_{th}</i>) and the rated dynamic current ix) Class of Insulation x) Rated continuous thermal current xi) Property Label – " Property of WBSEDCL" xii) Guaranteed for 5 Years
5.0	Voltage Transformers	
5.1.	Type	Dry / Resin cast, draw out type, 3-phase, 5-limb or it should be combination of three no Single Phase PT housed in a withdrawable carriage complete with HV fuses. Copies of Type test certificate and routine test certificate as per IEC 60044-2, IS 16227 (PART 1 & 3) shall be submitted with the tender for evaluation purpose. Each PT shall be provided with Vector diagram plate as per relevant provision of IS showing primary and secondary terminals.
5.2.	Rated Voltage	
5.2.1	Primary	11000 /√3V
5.2.2	Secondary	110 /√3V
5.3	No of phases	3
5.4	No. of secondary windings	1(one)
5.5	Method of connection	Star/Star with both neutral earthed
5.6	Rated voltage factor	1.2 continuous, 1.9 for 8 hours.
5.7	Class of insulation	Class E or better
5.8	Accuracy class	0.5
5.9	Primary and secondary fuses	HRC current limiting type, primary fuse Replacement shall be possible with VT in withdrawn position
5.10	Voltage class	12 kV
5.11	VA Burden	100VA per phase



5.12	PT mounting	Rail mounted on top of the Incomer Panel and Primary connected on bus side. It can be plugged into and withdraw from service by pulling or pushing the PT by the handle provided on the PT. This action traverses the PT along the rails and shall automatically operate the spout shutters. The shutter drive also forms a latch which holds the PT in the service position and this latch shall be required to be released before PT can be isolated.
5.13	PT secondary circuit	The secondary circuits from all the PTs of the incoming panels shall be wired up to all the outgoing panels so that supply to the potential coils of the meters can be maintained from any of the PTs. Plug and socket type contacts shall be provided in the PT secondary. PT secondary terminals shall be inaccessible from outside. Necessary provision shall be made for sealing PT while in racked in position. Vector of the PT will be Both Y/Y connection with both neutral earthed.
5.14	Name Plate of PT indelibly marked	Name plate for the voltage transformer shall be provided with all the required details as per IEC 60044-2, IS 16227 (PART 1 & 3) Standards, including :- i) The manufacturer's name ii) Serial number and a type designation iii) Rated primary and secondary voltage iv) Rated frequency v) Rated output and corresponding accuracy class for each secondary winding, vi) The highest voltage of the equipment vii) The rated Insulation level ix) Class of Insulation for the equipment x) Rated voltage factor and corresponding rated time. xi) The use each secondary winding and it's corresponding terminals xii) Property Label – " Property of WBSEDCL" xiii) Guranteed for 5 Years
6.0	Monitoring from Remote	Suitable arrangement to be provided to monitor the following conditions of 11kV VCB at SCADA Control Centre end through installing RTU by WBSEDCL:- 1 . DC healthy 2 .Breaker is 'OFF' 3 .Breaker is 'ON' 4 .Spring for closing mechanism is charged 5 .Breaker is in Local 6 .Breaker is in Remote 7. Trip Circuit healthy 8 .Tripping due to O/C 9 .Tripping due to E/F

**Annexure-C****Legend of Devices associated with Multi panel Shunt trip vacuum Circuit Breaker**

Symbol Reference	Description
A1, A2, A3 & AH(heater)	Ammeter
V and V _{DC} for Bus coupler	Voltmeter
VS	Manual Voltmeter Selector Switch
EM	Energy Meter
CS	Control switch T-A / T-N-A / C-C spring return type
IL-R	CB 'ON' Indication Red lamp
IL-G	CB 'OFF' Indication Green lamp
IL-W	'Trip /Close signal received from Remote Indication white lamp
IL-B	"Spring charged" Indication Blue lamp
IL-A	CB " Auto trip" Indication Amber lamp
PB	Push Button
ANN	DC operated electric Annunciator
H,HS,TH	Heater, Heater Switch, Thermostat
FS	Fuse
LK	Link
MCB1	MCB, 2 pole 16 A for DC supply
MCB2	MCB, 2 pole 16 A for AC supply
MCB3	MCB, 2 pole for spring charging motor supply
MVS	Manual PT selector switch
IR-I	Remote inter tripping contact from 33 kV Transformer Control and relay Panel
TC	Tripping Coil
CC	Closing Coil
APR	Anti-pumping Relay
86	Tripping Relay
52	Vacuum Circuit breaker
52a,52b	NO and NC contacts of Breaker Auxiliary switch respectively
PT	Potential Transformer
CT	Current Transformer
TTB	Test Terminal Block
51/50 R-Y-B-N	O/C and E/F Relay

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Annexure D

Sl no.	Device	Quantity		
		Incomer	Feeder	Bus coupler
1.	a) Numerical O/C & E/F Relay(Directional)	1 No.	NIL	NIL
	b) Numerical O/C & E/F Relay		1 No.	1 No.
2.	Trip Relay	1 No.	1 No.	1 No.
3.	AC Supervision Relay	Nil	Nil	1 No.
4.	Annunciator	1 No.	1 No.	1 No.
5.	AC Ammeter (CT operated)	3 No.	3 No.	Nil
6.	AC Voltmeter	1 No.	Nil	Nil
7.	DC Voltmeter	Nil	Nil	1 No.
8.	Ammeter (Heater Ckt.)	1 No.	1 No.	1 No.
9.	Current Transformer	3 No.	3 No.	Nil
10.	Potential Transformer	1 Set	Nil	Nil
11.	Energy meter	1 No.	1 No.	Nil
12.	TTB	1 No.	1 No.	Nil
13.	Indicating Lamp	6 No.	6 No.	4 No.
14.	MCB (DC Ckt)	1 No.	Nil	1 No.
15.	MCB (AC Ckt, 16 Amps)	1 No.	Nil	1 No.
16.	MCB (Motor ckt.)	1 No.	1 No.	1 No.
17.	3 Position PT Selector Switch (16 Amps)	1 No.	1 No.	Nil
18.	Voltage Selector Switch	1 No.	Nil	Nil
19.	Electronic Buzzer (DC Volt operated)	Nil	Nil	1 No.
20.	Electronic Buzzer (AC Volt operated)	Nil	Nil	1 No.
21.	Heater with controller	1 Set	1 Set	1 Set
22.	T-N-C Switch	1 No.	1 No.	1 No.
23.	Local – Remote Switch	1 No.	1 No.	1 No.
24.	Mushroom type--Push Button for Emergency Trip	1 No.	1 No.	1 No.
25.	Push Button for Relay reset	1 No.	1 No.	Nil
26.	Spring Charging Handle	1 No. for every three Panel.		
27.	VCB Operating Handle	1 No. for every three Panel		
28.	HRC Fuse & Link	As required	As required	As required
29.	Trip Circuit supervision Relay	1 No.	1 No.	1 No.
30.	Push Button for Annunciator Accept & Reset	2 Nos.	2 Nos.	2 Nos.
31.	Push Button for Master trip relay reset.	1 No.	1 No.	1 No.
32.	DC Supervision Relay	Nil	Nil	1 No.

NB: The Bidder has to include list of other Items for the Equipment to meet our Technical requirement.

**Annexure-G****GURANTEED TECHNICAL PARTICULARS**

(To be submitted by the Bidder)

1.	General :	
	Name of the Company	
	Office address	
	Factory address	
2.	Panel	
	Make	
	Type & Designation	
	Application Standard	
	Rated Voltage	
	Highest Voltage	
	Normal Current	
	Frequency	
	STC for 3 Sec.	
	Breaking Capacity	
	Making Capacity	
	Power frequency withstand voltage	
	Impulse withstand voltage	
	AC Aux. Voltage.	
	DC Aux. Voltage	
	Degree of Protection for HV & LV Compartment	
	Thickness of metal sheet (mm).	
	For load-bearing member	

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	For Doors & covers	
	Dimension of Incomer Panel in mm (H x W x D)	
	Dimension of Outgoing Panel in mm (H x W x D)	
	Dimension of Bus-Coupler & Trunk Panel in mm (H x W x D)	
	Single feeder panels to be supplied with Bus Bar for connection with adjacent Incomer / Bus Coupler / Feeder. [Yes / No]	
	HV Cable termination height in the cable compartment	
3.	Bus Bar	
	Material	
	Shape	
	Cross sectional area of main Bus.	
	Cross sectional area of Riser.	
	Type of plating	
	Normal Current carrying capacity	
	STC for 3 Sec.	
	Temp. Rise over ambient at normal current	
	Current density for main Bus & Riser	
	Phase to Phase clearance	
	Phase to ground clearance	
	Type of insulation	
	Colour Identification	
	Bus Bars sleeve Full voltage sleeved with shrouds on joints	
4.	Bus support insulator	

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	Reference Standard.	
	Make	
	Material	
	Dry Power frequency Withstand Voltage for one minute	
	Wet Power frequency Withstand Voltage for one minute	
	Impulse Withstand voltage	
	Creepage distance	
5.	Vacuum Circuit Breaker	
	Make	
	Type	
	Reference Standard	
	Rated voltage	
	Highest voltage	
	Frequency	
	Normal Current	
	Breaking capacity	
	Making capacity	
	STC for 3 Sec.	
	Temp. Rise over ambient at normal current	
	No of break per phase.	
	Operating duty cycle	
	Single Phase Capacitor Breaking capacity	
	Three Phase Capacitor Breaking capacity	
	Line Charging Breaking capacity	

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	Cable Charging Breaking capacity	
	Closing time	
	Opening time	
	Mechanical Endurance capacity	
	Electrical Endurance capacity	
	Operating mechanism	
	Type of isolation & Drawout	
	Details of mechanical interlock provided	
	DC Wattage of Trip Coil.	
	DC Voltage range for Trip operation.	
	DC Wattage of Closing Coil.	
	DC Voltage range for Closing operation.	
	Details of electrical interlock.	
6.	Vacuum Bottle	
	Make	
	Rated voltage	
	Type and model no.	
	Normal current	
	Breaking capacity	
	Making capacity	
	STC for 3 Sec.	
	Maximum contact separation length	
	Minimum Mechanical life in no. of operation	
	Minimum Electrical Life in no. of operation at rated normal current	
	Minimum Electrical Life in no. of operation at rated	

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	full short circuit current	
	Power frequency withstand voltage (dry)	
	Impulse withstand voltage	
	Contact material	
	Type of plating	
	Contact pressure	
7.	Current Transformer	
	Make	
	Reference Standard	
	Type	
	Voltage	
	Frequency	
	Insulation level	
	Class of insulation	
	Ratio (Incomer & Feeder)	
	Class of accuracy (Incomer & Feeder)	
	Burden (Incomer & Feeder)	
	STC for 1 Sec.	
	ISF of Metering Core at lower ratio	
	Vk & Magnetising Current(at lower ratio)	
	Fixing Arrangement	
8.	Potential Transformer	
	Make	
	Reference Standard	
	Insulation level	

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	Type	
	Size	
	Basic CTR	
	Accuracy class	
	AC Aux. voltage	
	Fixing type.	
12.	Digital Ammeter (Heater circuit)	
	Make	
	Type	
	Size	
	Current range	
	Accuracy class	
	AC Aux. voltage	
	Fixing type.	
13.	Digital Volt meter (PT operated)	
	Make	
	Type	
	Size	
	PTR	
	Accuracy class	
	AC Aux. voltage	
	Fixing type.	
14.	Analog Voltmeter (DC Circuit)	
	Make	
	Type	

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	Size	
	Accuracy class	
	Fixing type.	
15.	Energy meter	
	Make	
	Type	
	Model no.	
16.	Terminal connector	
	Make	
	Type	
	Size	
17.	Control Switch	
	Make	
	Type	
	Contact configuration	
	AC & DC Current rating	
	AC Voltage rating.	
18.	Spring Charge Limit Switch	
	Make	
	Type	
	No of contact (NO + NC)	
	No of spare contact (NO + NC)	
	AC & DC Current rating	
	AC Voltage rating.	
19.	Local Remote Switch.	

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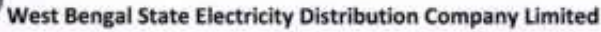
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	Make	
	Type	
	Contact configuration	
	AC & DC Current rating	
	AC Voltage rating.	
20.	Spring charging motor	
	Make	
	Type	
	Rated AC Voltage	
	Rated wattage	
	AC Voltage variation for operation	
21.	Operation counter	
	Make	
	Type	
	Operation count range	
22.	Space heater	
	Make	
	Type	
	Rated AC Voltage	
	Wattage	
23.	Thermostat	
	Make	
	Setting range	
24.	Indication Lamp	
	Make	

Technical Specification for 11 kV SCADA compatible Shunt trip switchgear panel

Page 47 of 70

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	Type	
25.	Auxiliary contact switch	
	Make	
	Type	
	No of contact (NO + NC)	
	No of spare contact (NO + NC)	
	AC & DC Current rating	
	AC Voltage rating.	
26.	Annunciator	
	Make	
	No of window.	
	Aux voltage	
	No of inbuilt push switch.	
27	Voltage selector switch	
	Make	
	Type	
28	PT selector switch	
	Make	
	Type	
	Current Rating	
29	TTB	
	Make	
	Type	
	Size	
30	Terminal Block	

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4	Information about submission of following documents with the Tender	
a)	Copies of Type Test Reports.	
b)	Copies of Tender Drawing Literature & Manual.	
c)	Copies of Tender Performance Reports & Credential	
d)	Copies of Tender Literature & Leaflets of Device	
e)	List of Recommended Spares.	
5	Information about submission of Deviation Sheet	

**ANNEXURE-H****GTP for Numerical Feeder (Non directional) Protection Relay**

Sl. No.	Feature and Function	Bidders response
1.1	Make	
	Type	
	ORDERING CODE (with supporting data sheet)	
1.2	Conformance to i. IEC255-4	
	ii. IEC 61850	
1.3	No. of CT inputs for O/C and E/F Protection	
1.4	Type test report submitted(y/n)	
1.5	Relay shall be of Numeric Design	
1.6	Relay designed for bay protection and Control	
1.7	Size of Relay LCD screen	
1.8	Relay is equipped with CB close and open key/push buttons	
1.9	Relay has following protection functions: a. Three phase over current b. Earth fault c. Thermal overload function d. Broken conductor protection function e. Circuit Breaker Maintenance function	
2.	a. One time delayed element and two high set elements b. Setting range and step for IDMT element for both current and Time Multiplier Setting c. Selectable Current/Time Curve for IDMT element d. Setting range and step for high set elements for both current and time delay	
3.	Sampling rate and frequency of analog signal	
4.	Whether remote controllable from SCADA	
5.	a. No. of Digital Inputs b. Voltage rating of Digital Inputs c. Provision of testing without current injection	

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6.	Supervision for CB open and Closed status	
7.	No. of programmable LEDs and no. of Latched LEDs	
8.	Analog Measurement and display supported	
9.	Fault Record storage capacity	
10.	Event storage capacity	
11.	Disturbance record storage capacity	
12.	MMI with keypad and LCD provided	
13.	Rated DC Supply and tolerance	
14.	Rating of CT/PT secondary	
15.	Rated frequency	
16.	a. Operating ambient temperature & humidity	
	b. Withstanding capability of Electromagnetic interference as per relevant part of IEC 61850	
17.	Mounting	
18.	Watchdog	
19.	a. Nominal Feeder current	
	b. CT Ratio setting	
	c. Earth fault current with time delay IEC Curves, 2 nd stage for instantaneous trip(less than 50 ms)	
	d. High set with delay	
	e. IEC Curves for all O/C and E/F have user selectable?	
20.	a. No. of Digital Output Contacts	
	b. Contact rating	
21.	Mode of Time Synchronization	
22.	Type of Lugs and terminators	
23.	MTBF	
24.	Lifespan	
25.	Compliance to Type Test	
26.	Communication Port a. Rear port- details b. Front port-details	
27.	Whether Communication Ports are native to the Relay	
28.	Protocol supported for Rear Port	



29.	Protocol supported for Front port	
30.	Start and trip output contacts are freely programmable	
31.	Cable for connection of Relay to laptop(USB port) along with converter and power supply if required for relay local setting	
32.	Basic application software for setting change, parameterisation	
33.	CD with software(licensed) to download disturbance recorder, event log and evaluation of those records	
34.	Graphical configuration tool for I/P, O/P and functional building block for protection and control	
35.	Any other software required for integration with SCADA.	

ANNEXURE-J**GTP for Master Trip Relay**

Sl. No.	Description	Bidder's Response
01.	Manufacturer Name	
02.	Type and designation	
03.	ORDERING CODE (with supporting data sheet)	
04.	Electrical reset	
05.	Mounting	
06.	High Burden relay	
07.	Operating Time	
08.	Rated DC supply and tolerance	
09.	No. of NO Contact	
10.	No. of NC Contact	

Date :

Signature:

Name:

Designation:

Company Seal:

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**ANNEXURE-E**

All numerical relays shall be cyber security compliant; in line with the guidelines of the Ministry of power, Govt. of India.

Technical Specification for Non-directional over current and earth fault relay

Sl. No.	Feature and Function	Technical requirement
1	Purpose and application	It is intended to automate the Switchgears specified in the scope of supply and use Communicable Numeric relays for Protection, Control, Metering and Status monitoring. This specification is based on the understanding that an integrated Automation System along with protections shall be provided and same shall have provisions for Integration with SCADA system. All the feeders shall be remote controlled from Employer's SCADA and from the local console of the numerical relays. Numerical multifunctional combined Microprocessor based Feeder protection and management relay to protect the 11 kV Feeder from all electrical and other faults along with reporting system, Disturbance record for fault analysis. Bidder should comply with any especial requirement or feature asked for retrofitting the relays. Relay should be IEC 61850 compliant. Relay should have 4 CT input for O/C and E/F protection. There should be option for derivation of E/F internally.
2.	Main Protection Feature	- Relay should have minimum two group of setting. Setting group changeover required from digital status input. - Electrical over load protection with selectable IEC curves with two stage, first stage to be used as Definite Time / IDMT and second stage to be used as high set for short circuit protection. - Earth fault protection in two stages with IEC characteristics. First stage to be used as IDMT/Definite Time and second stage to be used as instantaneous elements. Earth fault element should be suitable for both CBCT and residual type CT connection. - Negative phase sequence Protection with IEC Curves. - CB Fail Protection & time settable as per user. - The relay should be immune to DC switching while carrying current i.e. no spurious trip should be generated if relay DC is made On and Off - The relay should conform to the IEC255-4 or BS 142 for Inverse time characteristics. - The relay should have features to monitor for broken conductor and CB opening time.
3.	Processor feature	Relay shall be completely Numerical with protective elements having software algorithm based on sampling of analog inputs. Sampling Rate of Analog Signal: The sampling rate should be 1000 Hz for 50 Hz signal or better for each analog channel. Hardware based

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		measurements shall not be acceptable.
4.	Operational Philosophy	The operation of Relay shall be possible both locally from the Switchgear and remote & Local Work station. The local position shall be displayed in remote / local workstation and remote operation shall be blocked if the switch is in Local. Clear control priorities shall prevent initiation of operation of a single switch at the same time from more than one of the various control levels and there shall be interlocks among various control levels. The priority shall always be with the lowest enabled control level. Relay accuracy shall not be affected by system frequency fluctuation.
5.	Status/Optical Inputs/Digital inputs	1. Minimum 10 number status inputs are required 2. All status inputs should be 30 V DC. 3. Setting group is required to be changed with any Digital input status. 4. Trip circuit supervision with DI status 5. The digital inputs shall be acquired by exception with 1ms resolution. Contact bouncing in digital inputs shall not be assumed as change of state. 6. Relay should have comprehensive self diagnostic feature with remote indication of relay failure and alarm shall be generated without tripping of circuit 7. Provision of Testing output relays without any current injection. 8. No. of programmable LEDs- at least 8nos. with latching option.
6.	Main measuring and reporting feature	All measurements should be in primary quantities. Minimum following displays are required in alpha numeric:- 1. Three phase (Positive sequence) current 2. Neutral(zero sequence) current 3. All the trips should have clear indication on the relay terminals 4. Resetting should be selectable as hand reset or auto reset. 5. The default relay LCD shall be user defined to display primary circuit loading.
7.	Memory and Recording Feature	1. The relay setting and programming should be stored in EEPROM so that during Aux. Power failure the said data is not lost. 2. Relay should have event log, trip log and DR record. All logs should go in to history. 3. All tripping of relay should initiate DR in auto without extra binary input. Triggering of DR with binary input should be user configurable. 4. The last 2 fault DR records should be in flash memory and DR will not erase in case of DC supply fail for more than 2 days. 5. Should be able to record at least 5 Oscillographic disturbances and 5 fault records and 250 event records. 6. Minimum Four no. of latest trip log with cause of trip should be stored in memory along with date and time stamping. The memory should not be lost with the switching off of DC. 7. The relay should have fault-recording feature with current waveform and Digital Input status. The fault waveform should consist of minimum four current waveforms of three phases current and zero sequence current and DI status. Triggering time for Pre and Post should have user

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		selectable. This record should be in flash memory for minimum 7(seven) days even after switching off the DC supply. 8. The fault should be date and time stamped. 9. Communication protocol IEC 61850.
8.	Auxiliary Supply	30 V DC to - 25% to + 10%, 2 wire unearthed system. Necessary software shall be in-built for proper shutdown and restart in case of power failure. Auxiliary supply burden will be around 20Watt.
9.	Rated CT/PT secondary	5/1 Amp(User selectable) , CTs used to be protection class
10.	Rated frequency	50 HZ +/- 3%
11.	Ambient condition	1. Operating ambient temperature upto 55 Deg C 2. Operating Humidity upto 100 % 3. Relay shall meet the requirement for withstanding electromagnetic interference according to relevant parts of IEC 61850. Failure of single component within the equipment shall neither cause unwanted operation nor lead to a complete system breakdown.
12.	Module and Mounting	1. Relay should be flush mounted type 2. If module is draw-out type then it should have CT shorting facility of make before break type. 3. Mounting in switchgears located in non AC rooms. 4. Galvanic isolation between field connection and relay hardware should be there.
13.	Watchdog and self monitoring	The relay should have facility to monitor the healthiness of its circuits and components by own monitoring system. In case of any problems, the alarm should be generated by one of the output contacts. The alarm as soft signal to be sent to SCADA system as well. Necessary support documentation explaining the self diagnostic feature shall be furnished.
14.	Relay parameterisation and Settings	Settings possible through relay keypad relay HMI software should be as follows:- 1. Nominal Feeder current 2% to 110 % 2. CT Ratio setting 10-1000(approx.) 3. Earth fault current 5 to 40% with time delay IEC Curves, 2nd stage for instantaneous trip(less than 50 ms) 4. Over current trip- 50% to 200% of 1/5 Amp with time delay as per IEC Curves. 5. High set with delay 200% to 2000% 6. IEC Curves for all O/C and E/F have user selectable.
15.	Output Relays	Minimum 10number output relays are required out of which 1. One potential free change over contact should be provided for start inhibit of relay. 2. All o/p contact should be freely programmable. 3. Rating of trip contacts:- a) Contact durability>10K operation b) 15 Amp make and carFry for 3 sec for trip contact c) Make and carry for trip contacts L/R<=40ms 4. Rating of Alarm contacts:- 5 Amp make and carry continuously for 5 sec. Testing of Output relays through keypad on relay fascia and relay HMI software. Output relay dwell time shall be user programmable or fixed at 100ms.

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16.	Relay software and Man Machine Interface	1. The relay should have native IEC 61850 Communication Protocol. 2. Should have password protected key padlock. 3. Necessary software for relay setting , retrieving DR, event log, trip log should be supplied by the Bidder. Necessary License is to be issued for WBSEDCL, if required. 4. Bidder has to supply communication hardware for relay setting, DR downloading from front port. This device should be compatible to USB/Ethernet port. 5. It shall be possible to transfer the data stored in the DFR to computer on IEEE/COMTRADE format. The data format shall be compatible for dynamic protection relay testing on relay test kit. COMTRADE data viewing software to be provided. 6. Multiuser/Corporate license for installation on minimum 7 nos. of PCs.
17.	Date and time	Date and Time stamping with faults and record. The clock should be powered from internal cell and should not required setting after every DC switching. The internal cell life minimum 5 years. Time synchronization by IRIG-B or SNTP. For time synchronization through SNTP is to be provided from clock signal coming from RTU. In case of IRIG-B, time synchronization will be done with GPS clock signal from GPS receiver located at substation.
18.	Lugs and terminators	All CT and PT terminals shall be provided as fixed (screwed) type terminals on the relay to avoid anyhazard due to loose connection leading to CT opening or any other loose connection. All CT terminals of the Relay to be compatible with ring type lug. Necessary amount of lugs should be supplied along with each relay for CT connection and control wiring.
19.	Manuals, Drawings and Literature	1. The relays should be supplied with manuals with all technical and operating instructions. 2. All the internal drawings indicating the logics and block diagram details explaining principle of operation should be given at the time of supply. 3. Mapping details shall be submitted in IEC format.
20.	Assistance in commissioning	As part of this offer, Bidder shall have to provide necessary support during integration of relays with the third party SCADA system which will be commissioned in future.
21.	Standard documentation per Relay, according to IEC 61850	1. MICS document(model implementation conformance statement) 2. PICS(protocol implementation conformance statement) 3. PIXIT document 4. Conformance Test certificate by KEMA/CPRI. All the above mentioned certificates shall be submitted along with Techno-commercial Bid. 5. ICD file 6. SCD file
22.	Extendibility in Future	The Bidder shall provide all necessary software tools along with source codes to perform addition of bays in future and complete integration with SCADA by the User. These software tools shall be able to configure relay, add analog variable, alarm list, event list, modify interlocking logics etc. for additional



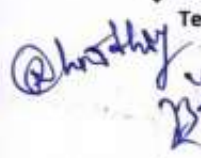
		bays/equipments which shall be added in future.
23.	Lifespan	The supplier should mention following:- 1. Product maturity: The Bidder should mention the time period for which the product is in the market 2. Expected production life 3. Hardware/Firmware change notification process. Upgrades to be provided free of cost within the warranty period/5 years whichever is later, if needed. 4. Lifespan of standard tools and processes for relay configuration, querying and integration.
24.	Standards	The relay should conform to the IEC255-5 or equivalent BS / ANSI for following:- 1. Overload withstand test 2. Dielectric withstand: 2kV in common, 1 kV in differential mode 3. Impulse Voltage: 5kV in common, 1kV in differential mode 4. Insulation resistance>100 M ohm. 5. Vibration: Shock and bump and Seismic 6. Storing and transportation 7. Radio Interference: IEC 61000 for high frequency disturbance, Transient disturbance, Electrostatic discharge 8. KEMA/CPRI Certification for the particular model offered with respect to IEC61850 Protocol.
25.	Communication Port	6. Two nos. IEC 61850 (Edition-2) protocol compliant 100 MBPS Ethernet RJ45/F.O ports having selectable HSR (High Availability Seamless Redundancy) and PRP (Parallel Redundancy Protocol) Protocol compliance for communication with SCADA system through two managed Ethernet Switches operating in redundant mode. The communication shall be made in 1+1 mode between individual IED to Switch, such that failure of one set of LAN shall not affect the normal operation of SCADA. However, it shall be alarmed in SCADA. The relay rear ports should accommodate both daisy chain and star bus topology for SCADA integration. 1. Functioning of Relay shall not hamper to fault occurring any interconnected relay. 2. One Front port Ethernet RJ45/USB 2.0 for relay parameterization and configuration etc. with the help of PC. 3. Relay should generate GOOSE message as per IEC 61850 standard for interlocking and also ensure interoperability with third party relays.
26.	Name Plate and marking	Each IED shall be clearly marked with manufacturer's Name, type, serial no. and electrical rating data. Name plates shall be made of anodized aluminium with white engraving on black surface.

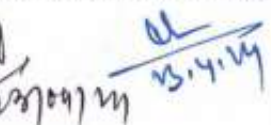
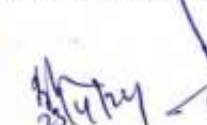
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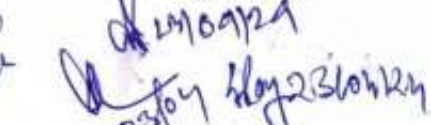
Page 59 of 70

**Technical specification for IEC 61850 compliant Directional O/C and E/F Relay**

Sl. No.	Feature and Function	Technical requirement
1.	Purpose and application	It is intended to automate the Switchgears specified in the scope of supply and use Communicable Numeric relays for Protection, Control, Metering and Status monitoring. This specification is based on the understanding that an integrated Automation System along with protections shall be provided and same shall have provisions for Integration with SCADA system. All the feeders shall be remote controlled from WBSEDCL's SCADA and from the local console of the numerical relays. Numerical multifunctional combined Microprocessor based Feeder protection and management relay to protect the 33 kV Feeder from all electrical and other faults along with reporting system, Disturbance record for fault analysis. Bidder should comply with any especial requirement or feature asked for retrofitting the relays. Relay should be IEC 61850 compliant. Relay should have 4 CT input, 3 input for O/C and residual E/F protection will be derived internally. One CT input may be used for unbalanced current protection. Relay should have 4 voltage input, 3 input for VT element for directional O/C protection with internally derived residual voltage for E/F protection. Another VT input will be used for residual voltage protection. Relay should have two stage over voltage and under voltage protection. The Directional O/C & E/F Relay should have the Site-selectable provision for operating in both Directional & Non-directional mode depending on site requirement.
2.	Main Protection Feature for directional O/C & E/F relay.	1. Electrical over load protection with selectable IEC curves with two stage, first stage to be used as Definite Time / IDMT and second stage to be used as high set for short circuit protection. 2. Earth fault protection in two stages with IEC characteristics. First stage to be used as IDMT/Definite Time and second stage to be used as instantaneous elements. Earth fault element should be suitable for both CBCT and residual type CT connection. 3. Negative phase sequence Protection with IEC Curves. 4. CB Fail Protection & time settable as per user. 5. The relay should be immune to DC switching while carrying current i.e. no spurious trip should be generated if relay DC is made On and Off 6. The relay should conform to the IEC255-4 or BS 142 for Inverse time characteristics. 7. VT fuse fail detection on NPS current/NPS Voltage or zero sequence current/voltage based logic and blocking of under voltage protection by VT fuse fail detection.





Technical Specification for 11 kV SCADA compatible Shunt trip switchgear panel

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Technical Specification for 11 kV SCADA compatible Shunt trip switchgear panel

Page 62 of 70
23109/24
23109 23109/24

Page 63 of 70



		WBSEDCL, if required. 4. Bidder has to supply communication hardware for relay setting, DR downloading from front port. This device should be compatible to USB/Ethernet port. 5. It shall be possible to transfer the data stored in the DFR to computer on IEEE/COMTRADE format. The data format shall be compatible for dynamic protection relay testing on relay test kit. COMTRADE Data viewer software is to be provided. 6. Multiuser/Corporate license for installation on minimum 7 nos. of PCs.
17.	Date and time	Date and Time stamping with faults and record. The clock should be powered from internal cell and should not required setting after every DC switching. The internal cell life minimum 5 years. Time synchronization by IRIG-B or SNTP. For time synchronization through SNTP is to be provided from clock signal coming from RTU. In case of IRIG-B, time synchronization will be done with GPS clock signal from GPS receiver located at substation.
18.	Lugs and terminators	All CT and PT terminals shall be provided as fixed (screwed) type terminals on the relay to avoid any hazard due to loose connection leading to CT opening or any other loose connection. Necessary amount of lugs should be supplied along with each relay for CT connection and control wiring. All CT terminals should be capable to connect with ring socket.
19.	Manuals, Drawings and Literature	- The relays should be supplied with manuals with all technical and operating instructions. - All the internal drawings indicating the logics and block diagram details explaining principle of operation should be given at the time of supply. - Mapping details shall be submitted in IEC format.
20.	Standard documentation per Relay, according to IEC 61850	- ICS document (model implementation conformance statement) - PICS(protocol implementation conformance statement) - Conformance Test certificate from KEMA/CPRI. - PIXIT document All the above mentioned certificates shall be submitted along with Techno-commercial Bid. - ICD file - SCD file
21.	Extendibility in Future	The Bidder shall provide all necessary software tools along with source codes to perform addition of bays in future and complete integration with SCADA by the User. These software

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		tools shall be able to configure relay, add analog variable, alarm list, event list, modify interlocking logics etc. for additional bays/ equipments which shall be added in future.
22.	Lifespan	The supplier should mention following:- 1. Product maturity: The Bidder should mention the time period for which the product is in the market 2. Expected production life 3. Hardware/Firmware change notification process. Upgrades to be provided free of cost within the Guarantee period/5 years whichever is later, if needed. 4. Lifespan of standard tools and processes for relay configuration, querying and integration.
23.	Standards	The relay should conform to the IEC255-5 or equivalent BS / ANSI for following:- 1. Overload withstand test 2. Dielectric withstand: 2kV in common, 1 kV in differential mode 3. Impulse Voltage: 5kV in common, 1kV in differential mode 4. Insulation resistance>100 M ohm 5. Vibration: Shock and bump and Seismic 6. Storing and transportation 7. Radio Interference: IEC 61000 for high frequency disturbance, Transient disturbance, Electrostatic discharge 8. KEMA/CPRI Certification for the particular model offered with respect to IEC61850 Protocol
24.	Communication Port	1. Two nos. IEC 61850 (Edition-2) protocol compliant 100 MBPS Ethernet RJ45/F.O ports having selectable HSR (High Availability Seamless Redundancy) and PRP (Parallel Redundancy Protocol) Protocol compliance for communication with SCADA system through two managed Ethernet Switches operating in redundant mode. The communication shall be made in 1+1 mode between individual IED to Switch, such that failure of one set of LAN shall not affect the normal operation of SCADA. However, it shall be alarmed in SCADA. The relay rear ports should accommodate both daisy chain and star bus topology for SCADA integration. 2. Functioning of Relay shall not hamper to fault occurring any interconnected relay. 3. One Front port Ethernet RJ45/USB 2.0 for relay parameterization and configuration etc. with the help of PC. In case RS-232 port offered, suitable interfacing cable



		with one end having RS 232 port and other end USB 2.0 to be provided to connect with PC free of cost. 4. Relay should generate GOOSE message as per IEC 61850 standard for interlocking and also ensure interoperability with third party relays.
25.	Name Plate and marking	Each IED shall be clearly marked with manufacturer's Name, type, serial no. and electrical rating data. Name plates shall be made of anodized aluminium with white engraving on black surface.
26.	Performance Guarantee	Relays will be guaranteed for the period of five years from the date of last dispatch. Any problem in the said period should be attended free of charge inclusive of repair/replacement of relays/ component (both H/W, S/W).
27.	Type Test	• Dielectric Withstand Test—IEC 60255-5 • High Voltage Impulse Test, class III --- IEC 60255-5(5kV peak, 1.2/50 micro Sec;3 Positive and 3 negative shots at interval of 5 Sec.) • DC Supply Interruption ---- IEC 60255-11 • AC Ripple on DC supply ---- IEC 60255-11 • Voltage Dips and Short Interruptions --- IEC 61000-4-11 • High frequency Disturbance ---- IEC 60255-22-1, Class III • Fast Transient Disturbance ---- IEC 60255-22-4, Class-IV • Surge withstand capability ---- IEEE/ANSI C 37.90.1(1989) • Degree of Protection • Electromagnetic compatibility • Mechanical stress/vibration test • Temperature withstand Type test reports for the above tests shall be submitted for the approval of WBSEDCL along with Tender. Wherever the above mentioned standards and IEC 61850 overlap, the latter will prevail.
28.	Credential as pre-requisite of Tender	1. Copies of performance certificate for two years successful operation as on the due date of bid opening for the offered relay in respect to implementation of IEC 61850 protocol to any SCADA/substation automation system from reputed Power Sector Utility in India shall have to be furnished along with the Bid. Copies of Purchase Orders and corresponding Delivery Challans /Stores Receipt vouchers/ Excise Duty Invoice, etc., i.e. Proof of Execution of the Purchase Orders. OR Successful testing and operation of minimum one year in

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		WBSEB/WBSEDCL/WBSETCL network.. 2. Documentary evidence for being manufacturers like registration Certificate issued by SSI/NSIC/Directorate of Industries/DGS&D, etc. for Qualifying requirement. 3. The bidder should have testing facilities of all functional tests or should have arrangement of all functional tests at government approved testing laboratories. <u>Inter-operability test:-</u> After fulfilment of the above Q.R. inter-operability test of the offered relay (other than Make & Model used in WBSEDCL) with the existing relay in WBSEDCL Network will be tested in WBSEDCL Distribution Testing Department, WBSEDCL for which due intimation for supply of sampled of offered relay will be given to the Bidder. The Bidder needs to submit the said relay to Distribution Testing Department, WBSEDCL within one week from the said intimation. The offered relay will only be accepted after fulfilment of above Q.R. &successful inter-operability test at WBSEDCL system.
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Checklist for Bill of Material for supply

Technical Specification for 11 kV SCADA compatible Shunt trip switchgear panel

Page 67 of 70

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Sl. No.	Material	
1.	Relay (Model No.)	Qty as per Tender
2.	Lugs suitable for current and control, wiring	Qty as per Tender X Number of TBs in relay + 20% extra.
3.	Cable for connection of Relay to laptop(USB port). Along with converter and power supply if required for relay local setting	10 set
4.	Manual, Hard copy in good quality paper properly bounded	10 set
5.	Copy of Type Test certificate along with manual	With offer
6.	Basic application software for setting change,	10 nos.
7.	CD with software(licensed) to download disturbance recorder, event log and evaluation of those records	10 nos.
8.	Graphical configuration tool for I/P, O/P and functional building block for protection	10 nos.
9.	Any other software required for integration with SCADA.	10 nos.

N.B All the above tools/ Software should be compatible to WINDOWS XP/WINDOWS NT/WINDOWS 7 and latest WINDOWS Operating System.

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**Undertaking from Relay manufacturer**

We hereby confirm that the protective relay(s) type

- i)
ii)
iii)

offered by us against your tender No. _____ through
M/s. _____ are in our current range of production. We
also confirm that these relays will not be phased out by us in the next 10
years from the date of supply. Necessary repairs/replacements if
necessary during this period will be made available by us directly to
WBSEDCL or through panel manufacturer. It is also confirm that as per
WBSEDCL technical specification we will provide 5 years guarantee of all
supplied relay directly to WBSEDCL or through panel manufacturer.

Name & Designation :

Company Seal :

We have offered our relay(s) type _____

- i)
ii)
iii)

to M/s. _____ against WBSEDCL's tender
no. _____.

In this connection we hereby confirm that we would be extending all
therequired technical support and back-up guarantee to WBSEDCL
and M/s. _____ for the above mentioned relay(s).

Name & Designation :

Company Seal :

Note: Copy of Schedule 1 shall also to be submitted during approval of
drawing.

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ANNEXURE-F

**STANDARD TECHNICAL SPECIFICATION FOR CT/PT
OPERATED DLMS COMPLIANT TRIVECTOR ENERGY
METER FOR 11KV FEEDER METERING UPLOADED
IN THE WBSEDCL WEBSITE TO BE FOLLOWED**

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